Frederick Wang

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Education

University of Toronto – M.Sc. Applied Computing (AI Concentration) **September 2026 – Present**
Sample Coursework: Neural Networks, Deep Learning, Imitation Learning for Robotics, Computational Linguistics

University of British Columbia – B.Sc. Computer Science and Statistics **September 2021 – May 2026**
Awards: Dean's List (2022-2026), Trek Excellence Scholarship (2022), J. Fred Muir Scholarship (2022)
Sample Coursework: Machine Learning, Relational Databases, Data Structures and Algorithms, Software Engineering, Computer Hardware and Operating Systems, Statistical Inference, Stochastic Processes

Skills

Languages: Python, SQL, R, Java, C++, C, JavaScript
ML/AI: PyTorch, scikit-learn, Hugging Face, RAG, MCP, Imitation Learning, Time-Series Forecasting, Bayesian Modelling
Data & Cloud: Spark, Databricks, Unity Catalog, Great Expectations, AWS (Bedrock, OpenSearch, S3), Azure, SQL Server
Tools: Git, CARLA, Next.js, Node.js, JupyterLab

Work Experience

Teaching Assistant – UofT Department of Computer Science **September 2026 – Present**
- Lead weekly CSC207 tutorials and labs on Java, object-oriented design, and Git-based team development

Data Engineer Intern – British Columbia Investment Management **May 2025 – December 2025**
- Won 1st in an internal hackathon with a schedule-optimization algorithm that improved scheduling efficiency by 25%
- Led production deployment of the algorithm, projected to unlock $20M in enterprise value for a portfolio company
- Engineered and deployed a production MCP server on Databricks Apps connecting Claude.ai to Unity Catalog, cutting company-wide ad hoc analysis time by 50%
- Added SQL Server Managed Instance support to open-source Great Expectations for Databricks data validation
- Architected a RAG knowledge base for natural-language LLM querying of unstructured enterprise data

Teaching Assistant – UBC Departments of Computer Science and Statistics **September 2024 – April 2026**
- Led weekly tutorials on ML models in R and mentored 100+ students through office hours and project evaluation
- Co-facilitated 400-level database lectures on query optimization, concurrency control, and crash recovery
- Provided feedback and evaluation on student data science projects across 4 terms

Data Analyst Intern – Providence Research **January 2024 – August 2024**
- Built supervised ML models in R predicting ankle surgery outcomes from patient risk factors and procedure type
- Cleaned, analyzed, and modelled ankle surgery outcome data in RStudio for statistical testing
- Built visualizations communicating outcome trends and model findings to the orthopaedic research team

Projects

Recoverability-Aware World Models for Safe Autonomous Driving **October 2026 – Present**
- Co-developing a recoverability-aware world-model objective in PyTorch that predicts whether a vehicle can still recover, distilled into a DAgger-trained policy to cut false "safe" judgments of prediction-only models
- Building a headless CARLA pipeline that mines unrecoverable scenes and generates counterfactual variants with NVIDIA Cosmos for recovery-labeled training data
- Designing an evaluation protocol comparing recovery prediction against prediction-only world-model baselines on held-out mined scenarios

Undercurrent: Foundation Model Commodity Forecasting **September 2026 – Present**
- Benchmarking Chronos-2 with native covariates against a hierarchical Bayesian driver model combined via Bayesian Predictive Synthesis, scoring leakage-safe forecasts on CRPS and calibration

GreenLife – AWS x UBC CIC Generative AI and Sustainability Hackathon – 1st Place **October 2024**
- Built a Next.js + Python RAG platform (AWS Bedrock, OpenSearch, S3) matching users to Canadian sustainability incentives